Lecture 7: One-Sample Location Problems

Instructor: Yichi Zhang

STAT-S520 Introduction to Statistics

Outline

General Setup

Assessing Normality

Normal One-Sample Location Problems

The Sign Test

Outline

General Setup

Assessing Normality

Normal One-Sample Location Problems

The Sign Test

General Setup

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ and we observe $x_1, \ldots, x_n$.

General Setup

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ and we observe $x_1, \ldots, x_n$.
- We are interested in the location $\theta = \psi(P)$ of P.

General Setup

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ and we observe $x_1, \ldots, x_n$.
- We are interested in the location $\theta = \psi(P)$ of P.
- Depending on the context, θ can be either the population mean or the population median.

Four specific problems

- Lecture 6: θ is population mean EX_1; no assumptions about P (except finite variance), n large;

Four specific problems

- Lecture 6: θ is population mean EX_1; no assumptions about P (except finite variance), n large;
- θ is population mean; no assumptions about n but $P = \mathrm{N}(\mu, \sigma^2)$;

Four specific problems

- Lecture 6: θ is population mean EX_1; no assumptions about P (except finite variance), n large;
- θ is population mean; no assumptions about n but $P = \mathrm{N}(\mu, \sigma^2)$;
- θ is population median; no assumptions about n or P (except X_i's are continuous).

Four specific problems

- Lecture 6: θ is population mean EX_1; no assumptions about P (except finite variance), n large;
- θ is population mean; no assumptions about n but $P = \mathrm{N}(\mu, \sigma^2)$;
- θ is population median; no assumptions about n or P (except X_i's are continuous).
- θ is center of symmetry; no assumptions about n, but P is symmetric and X_i's are continuous.

Note: For the second bullet point, we consider point estimation, hypothesis testing, and confidence interval estimation (set estimation) as the inference tasks of interest; For the third and the fourth bullet points, we will focus on hypothesis testing primarily.

Outline

General Setup

Assessing Normality

Normal One-Sample Location Problems

The Sign Test

Assessing Normality

- Section 10.1: Exact inference based on normal population

Assessing Normality

- Section 10.1: Exact inference based on normal population
- Section 10.2: Sign test and sample median; Normality assumption not needed

Assessing Normality

- Section 10.1: Exact inference based on normal population
- Section 10.2: Sign test and sample median; Normality assumption not needed
- Section 10.3: Wilcoxon signed rank test; Normality assumption not needed but assume that the population has a symmetric pdf

Assessing Normality

- Section 10.1 is partially based on the normality assumption on the population

Assessing Normality

- Section 10.1 is partially based on the normality assumption on the population
- If you are not certain that the population is normal, then it is safer to go back to Chapter 9.

Assessing Normality

- Section 10.1 is partially based on the normality assumption on the population
- If you are not certain that the population is normal, then it is safer to go back to Chapter 9.
- If there is strong evidence against the normality assumption, then don't do this Section 10.1! Go back to Chapter 9 instead and use CLT!

Assessing Normality

- Section 10.1 is partially based on the normality assumption on the population
- If you are not certain that the population is normal, then it is safer to go back to Chapter 9.
- If there is strong evidence against the normality assumption, then don't do this Section 10.1! Go back to Chapter 9 instead and use CLT!
- **Q:** How do we know whether or not the population is normal?

Assessing Normality

- Section 10.1 is partially based on the normality assumption on the population
- If you are not certain that the population is normal, then it is safer to go back to Chapter 9.
- If there is strong evidence against the normality assumption, then don't do this Section 10.1! Go back to Chapter 9 instead and use CLT!
- **Q:** How do we know whether or not the population is normal?
- **A:** Technically it is impossible. But there is hope to detect non-normality.

Assessing Normality

- Section 10.1 is partially based on the normality assumption on the population
- If you are not certain that the population is normal, then it is safer to go back to Chapter 9.
- If there is strong evidence against the normality assumption, then don't do this Section 10.1! Go back to Chapter 9 instead and use CLT!
- **Q:** How do we know whether or not the population is normal?
- **A:** Technically it is impossible. But there is hope to detect non-normality.
- **Qualitative methods:** quantile-quantile plot (QQ plot or normal probability plot) and kernel density estimate
- `R` **command:** `qqnorm()` and `density()` functions

Assessing Normality

- Section 10.1 is partially based on the normality assumption on the population
- If you are not certain that the population is normal, then it is safer to go back to Chapter 9.
- If there is strong evidence against the normality assumption, then don't do this Section 10.1! Go back to Chapter 9 instead and use CLT!
- **Q:** How do we know whether or not the population is normal?
- **A:** Technically it is impossible. But there is hope to detect non-normality.
- **Qualitative methods:** quantile-quantile plot (QQ plot or normal probability plot) and kernel density estimate
- `R` **command:** `qqnorm()` and `density()` functions
- **Qualitative methods are not decisive! Only for illustration!**

Assessing Normality

- Section 10.1 is partially based on the normality assumption on the population
- If you are not certain that the population is normal, then it is safer to go back to Chapter 9.
- If there is strong evidence against the normality assumption, then don't do this Section 10.1! Go back to Chapter 9 instead and use CLT!
- **Q:** How do we know whether or not the population is normal?
- **A:** Technically it is impossible. But there is hope to detect non-normality.
- **Qualitative methods:** quantile-quantile plot (QQ plot or normal probability plot) and kernel density estimate
- `R` **command:** `qqnorm()` and `density()` functions
- **Qualitative methods are not decisive! Only for illustration!**
- **Quantitative method:** Kolmogorov-Smirnov test (K-S test)
- `R` **command:** `ks.test()` function

Assessing Normality

KS test:

- Mathematically more involved: Compute the statistic

$$D_n = \sup_x |\widehat{F}_n(x) - \Phi(x)|,$$

- $\widehat{F}_n$ is the `ecdf` and Φ is the cdf of normal (`pnorm`)
- Works for large n
- There is some distributional theory for D_n that resembles the use of CLT for testing population mean

Assessing Normality

KS test:

- Mathematically more involved: Compute the statistic

$$D_n = \sup_x |\widehat{F}_n(x) - \Phi(x)|,$$

- $\widehat{F}_n$ is the `ecdf` and Φ is the cdf of normal (`pnorm`)
- Works for large n
- There is some distributional theory for D_n that resembles the use of CLT for testing population mean
- Hypothesis is $H_0 : P$ is normal against $H_1 : P$ is not normal

Assessing Normality

KS test:

- Mathematically more involved: Compute the statistic

$$D_n = \sup_x |\widehat{F}_n(x) - \Phi(x)|,$$

- $\widehat{F}_n$ is the `ecdf` and Φ is the cdf of normal (`pnorm`)
- Works for large n
- There is some distributional theory for D_n that resembles the use of CLT for testing population mean
- Hypothesis is $H_0 : P$ is normal against $H_1 : P$ is not normal
- `ks.test(x, "pnorm")` gives your a p-value of the test

Assessing Normality

KS test:

- Mathematically more involved: Compute the statistic

$$D_n = \sup_x |\widehat{F}_n(x) - \Phi(x)|,$$

- $\widehat{F}_n$ is the `ecdf` and Φ is the cdf of normal (`pnorm`)
- Works for large n
- There is some distributional theory for D_n that resembles the use of CLT for testing population mean
- Hypothesis is $H_0 : P$ is normal against $H_1 : P$ is not normal
- `ks.test(x, "pnorm")` gives your a p-value of the test
- If the p-value is small, then there is strong evidence against normality!

Assessing Normality

KS test:

- Mathematically more involved: Compute the statistic

$$D_n = \sup_x |\widehat{F}_n(x) - \Phi(x)|,$$

- $\widehat{F}_n$ is the `ecdf` and Φ is the cdf of normal (`pnorm`)
- Works for large n
- There is some distributional theory for D_n that resembles the use of CLT for testing population mean
- Hypothesis is $H_0 : P$ is normal against $H_1 : P$ is not normal
- `ks.test(x, "pnorm")` gives your a p-value of the test
- If the p-value is small, then there is strong evidence against normality!
- If the p-value is large, then there is no sign of non-normality

Assessing Normality

KS test:

- Mathematically more involved: Compute the statistic

$$D_n = \sup_x |\widehat{F}_n(x) - \Phi(x)|,$$

- $\widehat{F}_n$ is the `ecdf` and Φ is the cdf of normal (`pnorm`)
- Works for large n
- There is some distributional theory for D_n that resembles the use of CLT for testing population mean
- Hypothesis is $H_0 : P$ is normal against $H_1 : P$ is not normal
- `ks.test(x, "pnorm")` gives your a p-value of the test
- If the p-value is small, then there is strong evidence against normality!
- If the p-value is large, then there is no sign of non-normality
- Again, this does not say that the population must be normal!

Assessing Normality

KS test:

- Mathematically more involved: Compute the statistic

$$D_n = \sup_x |\widehat{F}_n(x) - \Phi(x)|,$$

- $\widehat{F}_n$ is the `ecdf` and Φ is the cdf of normal (`pnorm`)
- Works for large n
- There is some distributional theory for D_n that resembles the use of CLT for testing population mean
- Hypothesis is $H_0 : P$ is normal against $H_1 : P$ is not normal
- `ks.test(x, "pnorm")` gives your a p-value of the test
- If the p-value is small, then there is strong evidence against normality!
- If the p-value is large, then there is no sign of non-normality
- Again, this does not say that the population must be normal!

Outline

General Setup

Assessing Normality

Normal One-Sample Location Problems

The Sign Test

Normal One-Sample Location Problems

- **Normal random sample**: Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{N}(\mu, \sigma^2)$.

Normal One-Sample Location Problems

- **Normal random sample**: Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{N}(\mu, \sigma^2)$.
- μ is both the population mean and median.

Normal One-Sample Location Problems

- **Normal random sample**: Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{N}(\mu, \sigma^2)$.
- μ is both the population mean and median.
- **Distribution of the sample mean:** $\bar{X}_n \sim \mathrm{N}(\mu, \sigma^2/n)$.

Normal One-Sample Location Problems

- **Normal random sample**: Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{N}(\mu, \sigma^2)$.
- μ is both the population mean and median.
- **Distribution of the sample mean:** $\bar{X}_n \sim \mathrm{N}(\mu, \sigma^2/n)$.
- **Sample median:** Sample median $\widetilde{X}_n$ is such that

$$|\{X_i : X_i > \widetilde{X}_n\}| = |\{X_i : X_i < \widetilde{X}_n\}|.$$

Normal One-Sample Location Problems

- **Normal random sample**: Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{N}(\mu, \sigma^2)$.
- μ is both the population mean and median.
- **Distribution of the sample mean:** $\bar{X}_n \sim \mathrm{N}(\mu, \sigma^2/n)$.
- **Sample median:** Sample median $\widetilde{X}_n$ is such that

$$|\{X_i : X_i > \widetilde{X}_n\}| = |\{X_i : X_i < \widetilde{X}_n\}|.$$

- In R this can be computed using the `median` function.

Normal One-Sample Location Problems

- **Normal random sample**: Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{N}(\mu, \sigma^2)$.
- μ is both the population mean and median.
- **Distribution of the sample mean:** $\bar{X}_n \sim \mathrm{N}(\mu, \sigma^2/n)$.
- **Sample median:** Sample median $\widetilde{X}_n$ is such that

$$|\{X_i : X_i > \widetilde{X}_n\}| = |\{X_i : X_i < \widetilde{X}_n\}|.$$

- In R this can be computed using the `median` function.
- **Central Limit Theorem of the sample median**

$$\sqrt{n}(\widetilde{X}_n - \mu) \xrightarrow{\mathcal{L}} \text{A mean-zero normal}$$

Normal One-Sample Location Problems

- **Asymptotic Relative Efficiency:** The asymptotic relative efficiency (ARE) between the sample median and sample mean is the ratio between the reciprocals of the asymptotic variances, i.e., when n is big,

$$\mathsf{ARE}(\widetilde{X}_n, \bar{X}_n) = \frac{\mathrm{Var}(\tilde{X}_n)}{\mathrm{Var}(\bar{X}_n)} = \frac{2}{\pi} \approx 0.64.$$

Normal One-Sample Location Problems

- **Asymptotic Relative Efficiency:** The asymptotic relative efficiency (ARE) between the sample median and sample mean is the ratio between the reciprocals of the asymptotic variances, i.e., when n is big,

$$\text{ARE}(\widetilde{X}_n, \bar{X}_n) = \frac{\text{Var}(\tilde{X}_n)}{\text{Var}(\bar{X}_n)} = \frac{2}{\pi} \approx 0.64.$$

- Both $\tilde{X}_n, \bar{X}_n \to \mu$ as $n \to \infty$. But how accurate they could be depend on their variaces.
- **Implication of ARE:** Using the sample median to estimate μ from a large sample is equivalent to randomly discarding 36% of that sample and using the sample mean with the remaining 64%.

Hypothesis Testing

σ^2 **known:**

- If P is unknown and n is large then $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$ approximately follows $\mathrm{N}(0, 1)$;

Hypothesis Testing

σ^2 **known:**

- If P is unknown and n is large then $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$ approximately follows $\mathrm{N}(0,1)$;
- If $P = \mathrm{N}(\mu, \sigma^2)$ then $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$ exactly follows $\mathrm{N}(0,1)$ regardless of what n is;

Hypothesis Testing

σ^2 **known:**

- If P is unknown and n is large then $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$ approximately follows $\mathrm{N}(0,1)$;
- If $P = \mathrm{N}(\mu, \sigma^2)$ then $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$ exactly follows $\mathrm{N}(0,1)$ regardless of what n is;
- Use the same test as in Lecture 6

Hypothesis Testing

σ^2 **known:**

- If P is unknown and n is large then $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$ approximately follows $\mathrm{N}(0,1)$;
- If $P = \mathrm{N}(\mu, \sigma^2)$ then $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$ exactly follows $\mathrm{N}(0,1)$ regardless of what n is;
- Use the same test as in Lecture 6
- For one-sided test, the same procedures apply here even with small n

Hypothesis Testing

Example

Consider 16 measurements of the lengths of nails that are i.i.d. following $\mathrm{N}(\mu, \sigma^2)$. The following are the data:

$$2.14, 2.10, 2.13, 2.15, 2.13, 2.12, 2.13, 2.10,$$
$$2.15, 2.12, 2.14, 2.10, 2.13, 2.11, 2.14, 2.11.$$

Assume $\sigma = 0.01$. Consider $H_0 : \mu = 2.12$ against $H_1 : \mu \neq 2.12$ with $\alpha = 0.05$.

Hypothesis Testing

Example

Consider 16 measurements of the lengths of nails that are i.i.d. following $\mathrm{N}(\mu, \sigma^2)$. The following are the data:

$$2.14, 2.10, 2.13, 2.15, 2.13, 2.12, 2.13, 2.10,$$
$$2.15, 2.12, 2.14, 2.10, 2.13, 2.11, 2.14, 2.11.$$

Assume $\sigma = 0.01$. Consider $H_0 : \mu = 2.12$ against $H_1 : \mu \neq 2.12$ with $\alpha = 0.05$.

The sample mean is $\bar{x} = 2.125$. The observed value of the test statistic is

$$z = \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} = 2.$$

The p-value is `2 * pnorm(-abs(z))` $= 0.046 < \alpha$. Therefore we reject the null hypothesis $H_0 : \mu = 2.12$.

Hypothesis Testing

σ^2 **unknown:**

- χ^2 distribution: Let $Z_1, \ldots, Z_n$ be independent standard Gaussian random variables.
- Let $Z_1, \ldots, Z_n$ be independent $\mathcal{N}(0,1)$ random variables.
- Then the random variable

$$Y = Z_1^2 + \cdots + Z_n^2$$

  follows the **chi-squared distribution** with n degrees of freedom, written as

$$Y \sim \chi^2(n).$$

- A chi-squared distributed random variable is always non-negative.

Hypothesis Testing

σ^2 **unknown:**

- t distribution: Let $Z \sim \mathcal{N}(0,1)$ and $Y \sim \chi^2(n)$ be independent.
- Then the random variable

$$T = \frac{Z}{\sqrt{Y/n}}$$

  follows the (Student's) t distribution with n degrees of freedom, denoted by

$$T \sim t(n).$$

  `pt`.
- If $Z \sim \mathcal{N}(0,1)$ and $Y \sim \chi^2(14)$ are independent and

$$T = \frac{Z}{\sqrt{Y/14}},$$

  then

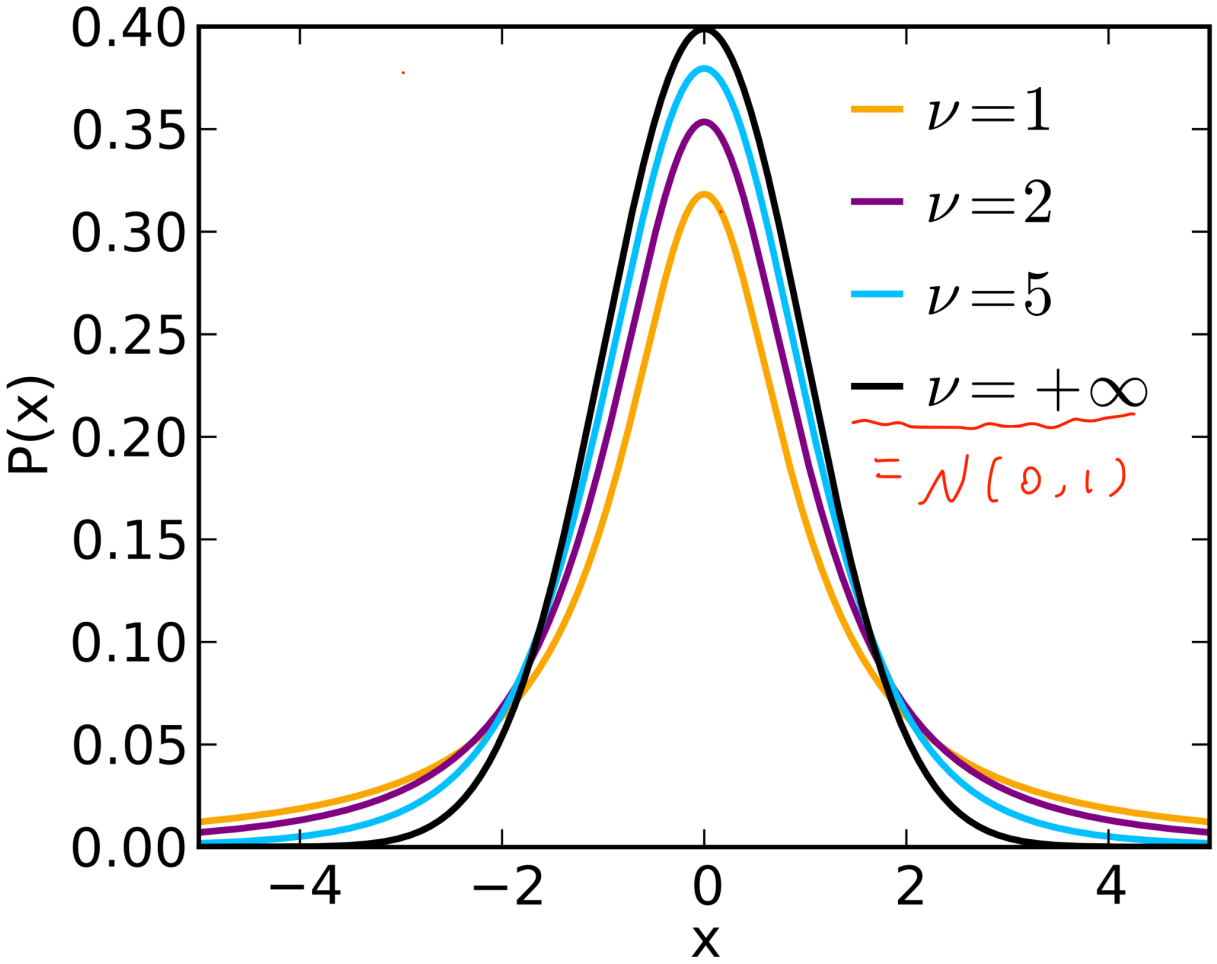
$$\mathbb{P}(T < -1.5) = \texttt{pt}(-1.5, \texttt{df} = 14) = 0.07791266.$$

- If $Y \sim t(n)$, then $\mathbb{E}Y = 0$.
- When n is large, $t(n)$ can be well approximated by $\mathcal{N}(0,1)$.

Hypothesis Testing

σ^2 **unknown:**

- When n is large, $t(n)$ can be well approximated by $\mathcal{N}(0,1)$.



Hypothesis Testing

σ^2 **unknown:**

- If P is unknown and n is large then $T_n = \sqrt{n}(\bar{X}_n - \mu)/S_n$ approximately follows $\mathrm{N}(0,1)$, where S_n^2 is the sample variance;

Hypothesis Testing

σ^2 **unknown:**

- If P is unknown and n is large then $T_n = \sqrt{n}(\bar{X}_n - \mu)/S_n$ approximately follows $\mathrm{N}(0,1)$, where S_n^2 is the sample variance;
- What if n is small? In that case, if we could believe $X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma^2)$, then our test statistics is

$$T_n = \sqrt{n}(\bar{X}_n - \mu)/S_n$$

exactly following a distribution regardless of what n is!

Hypothesis Testing

σ^2 **unknown:**

- If P is unknown and n is large then $T_n = \sqrt{n}(\bar{X}_n - \mu)/S_n$ approximately follows $\mathrm{N}(0,1)$, where S_n^2 is the sample variance;
- What if n is small? In that case, if we could believe $X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma^2)$, then our test statistics is

$$T_n = \sqrt{n}(\bar{X}_n - \mu)/S_n$$

  exactly following a distribution regardless of what n is!
- This is because $\bar{X}_n$ and S_n are independent random variables, and

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2(n-1)$$

- So we can write

$$T_n = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{S_n/\sigma} = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{\sqrt{(n-1)S_n^2/\sigma^2/(n-1)}} \sim t(n-1).$$

- **Student's 1-sample t-test:** We use a Students t distribution instead of the normal distribution to compute significance probabilities and critical values.

Hypothesis Testing

σ^2 **unknown:**

- If P is unknown and n is large then $T_n = \sqrt{n}(\bar{X}_n - \mu)/S_n$ approximately follows $\mathrm{N}(0,1)$, where S_n^2 is the sample variance;
- What if n is small? In that case, if we could believe $X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma^2)$, then our test statistics is

$$T_n = \sqrt{n}(\bar{X}_n - \mu)/S_n$$

  exactly following a distribution regardless of what n is!
- This is because $\bar{X}_n$ and S_n are independent random variables, and

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2(n-1)$$

- So we can write

$$T_n = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{S_n/\sigma} = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{\sqrt{(n-1)S_n^2/\sigma^2/(n-1)}} \sim t(n-1).$$

- **Student's 1-sample t-test:** We use a Students t distribution instead of the normal distribution to compute significance probabilities and critical values.

Hypothesis Testing

σ^2 **unknown:**

- If P is unknown and n is large then $T_n = \sqrt{n}(\bar{X}_n - \mu)/S_n$ approximately follows $\mathrm{N}(0,1)$, where S_n^2 is the sample variance;
- What if n is small? In that case, if we could believe $X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma^2)$, then our test statistics is

$$T_n = \sqrt{n}(\bar{X}_n - \mu)/S_n$$

  exactly following a distribution regardless of what n is!
- This is because $\bar{X}_n$ and S_n are independent random variables, and

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2(n-1)$$

- So we can write

$$T_n = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{S_n/\sigma} = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{\sqrt{(n-1)S_n^2/\sigma^2/(n-1)}} \sim t(n-1).$$

- **Student's 1-sample t-test:** We use a Students t distribution instead of the normal distribution to compute significance probabilities and critical values.
- One-sided test can be derived similarly

Hypothesis Testing

Example

Consider 16 measurements of the lengths of nails that are i.i.d. following $\mathrm{N}(\mu, \sigma^2)$. The following are the data:

$$2.14, 2.10, 2.13, 2.15, 2.13, 2.12, 2.13, 2.10,$$
$$2.15, 2.12, 2.14, 2.10, 2.13, 2.11, 2.14, 2.11.$$

Assume σ^2 is unknown. Consider $H_0 : \mu = 2.12$ against $H_1 : \mu \neq 2.12$ with $\alpha = 0.05$.

Hypothesis Testing

Example

Consider 16 measurements of the lengths of nails that are i.i.d. following $\mathrm{N}(\mu, \sigma^2)$. The following are the data:

$$2.14, 2.10, 2.13, 2.15, 2.13, 2.12, 2.13, 2.10,$$
$$2.15, 2.12, 2.14, 2.10, 2.13, 2.11, 2.14, 2.11.$$

Assume σ^2 is unknown. Consider $H_0 : \mu = 2.12$ against $H_1 : \mu \neq 2.12$ with $\alpha = 0.05$.

The sample mean is $\bar{x} = 2.125$ with sample variance $s^2 = 2.93 \times 10^{-4}$. The observed value of the test statistic is

$$t = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s} = 1.18$$

The p-value is 2`pt(-abs(t), df = 15)` $= 0.26 > \alpha$. Therefore we fail reject $H_0 : \mu = 2.12$.

Set Estimation

- σ^2 **known:** The confidence interval is

$$\bar{X}_n \pm q_z \frac{\sigma}{\sqrt{n}}, \quad \text{where} \quad q_z = \texttt{qnorm(1 - alpha/2);}$$

Example

Consider 16 measurements of the lengths of nails that are i.i.d. following $\mathrm{N}(\mu, \sigma^2)$. The following are the data:

$$2.14, 2.10, 2.13, 2.15, 2.13, 2.12, 2.13, 2.10,$$
$$2.15, 2.12, 2.14, 2.10, 2.13, 2.11, 2.14, 2.11.$$

Assume $\sigma = 0.01$. Consider the 90% confidence interval of μ.

Set Estimation

- σ^2 **known:** The confidence interval is

$$\bar{X}_n \pm q_z \frac{\sigma}{\sqrt{n}}, \quad \text{where} \quad q_z = \texttt{qnorm(1 - alpha/2);}$$

Example

Consider 16 measurements of the lengths of nails that are i.i.d. following $\mathrm{N}(\mu, \sigma^2)$. The following are the data:

$$2.14, 2.10, 2.13, 2.15, 2.13, 2.12, 2.13, 2.10,$$
$$2.15, 2.12, 2.14, 2.10, 2.13, 2.11, 2.14, 2.11.$$

Assume $\sigma = 0.01$. Consider the 90% confidence interval of μ.

The data leads to $\bar{x} = 2.125$ with $n = 16$. Assume that $\sigma = 0.01$. Then the 90% confidence interval is

$$2.125 \pm q_z \frac{0.01}{\sqrt{16}} = (2.1209, 2.1291), \quad \text{where} \quad q_z = \texttt{qnorm(1 - 0.1/2)}$$

Set Estimation

- σ^2 **unknown:** The confidence interval is

$$\bar{X}_n \pm q_{t(n-1)} \frac{S_n}{\sqrt{n}}, \quad \text{where} \quad q_{t(n-1)} = \texttt{qt(1 - alpha/2, df = n - 1);}$$

Set Estimation

- σ^2 **unknown:** The confidence interval is

$$\bar{X}_n \pm q_{t(n-1)} \frac{S_n}{\sqrt{n}}, \quad \text{where} \quad q_{t(n-1)} = \texttt{qt(1 - alpha/2, df = n - 1);}$$

- $t(n-1)$ has slightly larger variance than $\mathrm{N}(0,1)$

Set Estimation

- σ^2 **unknown:** The confidence interval is

$$\bar{X}_n \pm q_{t(n-1)} \frac{S_n}{\sqrt{n}}, \quad \text{where} \quad q_{t(n-1)} = \texttt{qt(1 - alpha/2, df = n - 1);}$$

- $t(n-1)$ has slightly larger variance than $\mathrm{N}(0,1)$
- $q_{t(n-1)}$ is slightly larger than q_z and the resulting interval is slightly wider

Set Estimation

- σ^2 **unknown:** The confidence interval is

$$\bar{X}_n \pm q_{t(n-1)} \frac{S_n}{\sqrt{n}}, \quad \text{where} \quad q_{t(n-1)} = \texttt{qt(1 - alpha/2, df = n - 1);}$$

- $t(n-1)$ has slightly larger variance than $\mathrm{N}(0,1)$
- $q_{t(n-1)}$ is slightly larger than q_z and the resulting interval is slightly wider
- Account for the extra uncertainty of not knowing the population variance.

Set Estimation

- σ^2 **unknown:** The confidence interval is

$$\bar{X}_n \pm q_{t(n-1)} \frac{S_n}{\sqrt{n}}, \quad \text{where} \quad q_{t(n-1)} = \texttt{qt(1 - alpha/2, df = n - 1);}$$

- $t(n-1)$ has slightly larger variance than $\mathrm{N}(0,1)$
- $q_{t(n-1)}$ is slightly larger than q_z and the resulting interval is slightly wider
- Account for the extra uncertainty of not knowing the population variance.

Example

Consider 16 measurements of the lengths of nails that are i.i.d. following $\mathrm{N}(\mu, \sigma^2)$. The following are the data:

$$2.14, 2.10, 2.13, 2.15, 2.13, 2.12, 2.13, 2.10,$$
$$2.15, 2.12, 2.14, 2.10, 2.13, 2.11, 2.14, 2.11.$$

Assume σ^2 unknown. Consider the 90% confidence interval of μ.

Set Estimation

Example

Consider 16 measurements of the lengths of nails that are i.i.d. following $\mathrm{N}(\mu, \sigma^2)$. The following are the data:

$$2.14, 2.10, 2.13, 2.15, 2.13, 2.12, 2.13, 2.10,$$
$$2.15, 2.12, 2.14, 2.10, 2.13, 2.11, 2.14, 2.11.$$

Assume σ^2 unknown. Consider the 90% confidence interval of μ.

Set Estimation

Example

Consider 16 measurements of the lengths of nails that are i.i.d. following $\mathrm{N}(\mu, \sigma^2)$. The following are the data:

$$2.14, 2.10, 2.13, 2.15, 2.13, 2.12, 2.13, 2.10,$$
$$2.15, 2.12, 2.14, 2.10, 2.13, 2.11, 2.14, 2.11.$$

Assume σ^2 unknown. Consider the 90% confidence interval of μ.

The data leads to $\bar{x} = 2.125$ with $n = 16$, and the sample variance is $s^2 = 2.93 \times 10^{-4}$. Then the 90% confidence interval is

$$2.125 \pm q_{t(15)} \frac{\sqrt{2.93 \times 10^{-4}}}{\sqrt{16}} = (2.1176, 2.1325),$$

where $q_{t(15)} =$ `qt(1 - 0.1/2, df = 15)`

Outline

General Setup

Assessing Normality

Normal One-Sample Location Problems

The Sign Test

The Sign Test

- **Problem setup:** Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ and we observe $x_1, \ldots, x_n$. Let θ be the population median: $P(X_1 < \theta) = 0.5$.

The Sign Test

- **Problem setup:** Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ and we observe $x_1, \ldots, x_n$. Let θ be the population median: $P(X_1 < \theta) = 0.5$.
- **Hypothesis testing:** We would like to test $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.

The Sign Test

- **Problem setup:** Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ and we observe $x_1, \ldots, x_n$. Let θ be the population median: $P(X_1 < \theta) = 0.5$.
- **Hypothesis testing:** We would like to test $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.
- **The sign test:**
  1. Let Y be the number of X_i's such that $X_i > \theta_0$. Y is the so-called test statistic; Let y be the number of x_i's such that $x_i > \theta_0$;

The Sign Test

- **Problem setup:** Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ and we observe $x_1, \ldots, x_n$. Let θ be the population median: $P(X_1 < \theta) = 0.5$.
- **Hypothesis testing:** We would like to test $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.
- **The sign test:**
  1. Let Y be the number of X_i's such that $X_i > \theta_0$. Y is the so-called test statistic; Let y be the number of x_i's such that $x_i > \theta_0$;
  2. $Y \sim \mathrm{Binom}(n, 0.5)$ under H_0;

The Sign Test

- **Problem setup:** Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ and we observe $x_1, \ldots, x_n$. Let θ be the population median: $P(X_1 < \theta) = 0.5$.
- **Hypothesis testing:** We would like to test $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.
- **The sign test:**
  1. Let Y be the number of X_i's such that $X_i > \theta_0$. Y is the so-called test statistic; Let y be the number of x_i's such that $x_i > \theta_0$;
  2. $Y \sim \mathrm{Binom}(n, 0.5)$ under H_0;
  3. Since $EY = n/2$ under H_0, the p-value is

     $$p = P_{\theta_0}\left(\left|Y - \frac{n}{2}\right| \geq \left|y - \frac{n}{2}\right|\right)$$

     and the sign test reject H_0 when $p \leq \alpha$, where α is the significance level.

The Sign Test

- **Problem setup:** Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ and we observe $x_1, \ldots, x_n$. Let θ be the population median: $P(X_1 < \theta) = 0.5$.
- **Hypothesis testing:** We would like to test $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.
- **The sign test:**
  1. Let Y be the number of X_i's such that $X_i > \theta_0$. Y is the so-called test statistic; Let y be the number of x_i's such that $x_i > \theta_0$;
  2. $Y \sim \mathrm{Binom}(n, 0.5)$ under H_0;
  3. Since $EY = n/2$ under H_0, the p-value is

     $$p = P_{\theta_0}\left(\left|Y - \frac{n}{2}\right| \geq \left|y - \frac{n}{2}\right|\right)$$

     and the sign test reject H_0 when $p \leq \alpha$, where α is the significance level.
  4. Take $c = \min(y, n - y)$
  5. Computation of the p-value:

     p-value $=$ `2 * pbinom(c, size = n, prob = 0.5)`.

The Sign Test

Derivation of the p-value:

- If $y \leq n/2$, i.e., $y \leq n - y$, then

$$\left|Y - \frac{n}{2}\right| \geq \left|y - \frac{n}{2}\right| \iff Y \geq n - y \text{ or } Y \leq y$$

and

$$\begin{aligned} p &= P_{\theta_0}(Y \leq y) + P_{\theta_0}(Y \geq n - y) \\ &= \texttt{2 * pbinom(y, size = n, prob = 0.5)} \end{aligned}$$

- If $y \geq n/2$, i.e., $y \geq n - y$, then

$$\left|Y - \frac{n}{2}\right| \geq \left|y - \frac{n}{2}\right| \iff Y \geq y \text{ or } Y \leq n - y$$

and

$$\begin{aligned} p &= P_{\theta_0}(Y \leq n - y) + P_{\theta_0}(Y \geq y) \\ &= \texttt{2 * pbinom(n - y, size = n, prob = 0.5)} \end{aligned}$$

The Sign Test

Example

consider the following sample of 20 observations on component lifetime (hr):

$$1.7, 3.3, 5.1, 6.9, 12.6, 14.4, 16.4,$$
$$24.6, 26.0, 26.5, 32.1, 37.4, 40.1, 40.5,$$
$$41.5, 72.4, 80.1, 86.4, 87.5, 100.2$$

Perform a sign test of H_0 : population median is 25 against H_1 : population median is not 25 at significance level $\alpha = 0.05$.

The Sign Test

Example

consider the following sample of 20 observations on component lifetime (hr):

$$1.7, 3.3, 5.1, 6.9, 12.6, 14.4, 16.4,$$
$$24.6, 26.0, 26.5, 32.1, 37.4, 40.1, 40.5,$$
$$41.5, 72.4, 80.1, 86.4, 87.5, 100.2$$

Perform a sign test of H_0 : population median is 25 against H_1 : population median is not 25 at significance level $\alpha = 0.05$.

```
n = length(x)
y = sum(x > 25)
c = min(y, n - y)
pvalue = 2 * pbinom(n - y, size = n, prob = 0.5)
pvalue
# [1] 0.5034447
```

Point Estimator from the Sign Test

- A point estimator can be derived from the sign test

Point Estimator from the Sign Test

- A point estimator can be derived from the sign test
- The sign test reject H_0 if $Y = \sum_{i=1}^{n} \mathbb{1}\left(X_i > \theta_0\right)$ deviates too large from $n/2$.

Point Estimator from the Sign Test

- A point estimator can be derived from the sign test
- The sign test reject H_0 if $Y = \sum_{i=1}^{n} \mathbb{1}(X_i > \theta_0)$ deviates too large from $n/2$.
- $H_0 : \theta = \theta_0$ would be true with the biggest chance if

$$Y = \sum_{i=1}^{n} \mathbb{1}(X_i > \theta_0) = n/2$$

- Namely

$$\sum_{i=1}^{n} \mathbb{1}(X_i > \theta_0) = \sum_{i=1}^{n} \mathbb{1}(X_i < \theta_0).$$

Point Estimator from the Sign Test

- A point estimator can be derived from the sign test
- The sign test reject H_0 if $Y = \sum_{i=1}^{n} \mathbb{1}(X_i > \theta_0)$ deviates too large from $n/2$.
- $H_0 : \theta = \theta_0$ would be true with the biggest chance if

$$Y = \sum_{i=1}^{n} \mathbb{1}(X_i > \theta_0) = n/2$$

- Namely

$$\sum_{i=1}^{n} \mathbb{1}(X_i > \theta_0) = \sum_{i=1}^{n} \mathbb{1}(X_i < \theta_0).$$

- Solve the equation for θ_0 gives an estimator

$$\widehat{\theta} = \text{sample median}$$

- **Confidence interval for the population median?**